

FL Simulation Results

File: test_A1

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 60.0% (0.60)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 60.0% (0.60) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8754
Clean + DP Protection Accuracy: 0.8761
Poisoned Accuracy: 0.8253
Poisoned + DP Protection Accuracy: 0.8261
Drop (Clean - Poisoned): 0.0501
Drop (Clean - Init): 0.0826
Drop (Clean DP - Init): 0.0833
Drop (Poisoned - Init): 0.0325
Drop (Poisoned DP - Init): 0.0333
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8820
Clean Recall: 0.8754
Clean F1 Score: 0.8748

Clean DP Precision: 0.8823
Clean DP Recall: 0.8761
Clean DP F1 Score: 0.8756

Poisoned Precision: 0.8348
Poisoned Recall: 0.8253
Poisoned F1 Score: 0.8245

DP Protection Precision: 0.8357
DP Protection Recall: 0.8261
DP Protection F1 Score: 0.8255

Summary

FL Simulation Complete
Task: ce315590-b516-4dac-b89a-ae7dd4c38d50
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8754
Clean DP Accuracy: 0.8761
Poisoned Accuracy: 0.8253
Data Poison Protection Accuracy: 0.8261
Drop (Clean - Poisoned): 0.0501
Drop (Clean - Init): 0.0826
Drop (Clean DP - Init): 0.0833
Drop (Poisoned - Init): 0.0325
Drop (Poisoned_DP - Init): 0.0333
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8820
Clean Recall: 0.8754
Clean F1 Score: 0.8748
Clean DP Precision: 0.8823
Clean DP Recall: 0.8761
Clean DP F1 Score: 0.8756
Poisoned Precision: 0.8348
Poisoned Recall: 0.8253
Poisoned F1 Score: 0.8245
DP Protection Precision: 0.8357
DP Protection Recall: 0.8261
DP Protection F1 Score: 0.8255